Classical quantum-state simulation. Reported calculations, sampled outcomes and represented model credits use separate units. Physical quantum gates executed: 0.

PDF generated UTC: 2026-10-10 19:13:34. Report renderer: 3.1.1-candidate. This is export time, separate from the original native calculation completion. Re-exporting archived results performs no new calculation. Visuals use returned native values.

Presentation `Detailed: readable meaning, returned fields and figures`

Export limits: 512 pages, 4 MiB of report text and 32 MiB of PDF bytes. Exports exceeding these limits are refused; use complete TXT/JSON or smaller command batches.

Retained command records `5`

These reports describe recorded responses and job snapshots. Exporting them does not execute another measurement or calculation. Complete raw native responses remain available through the JSON and TXT downloads.

q.input fit 67b43c4e300b4882af7de00409533dc0 target=temperature features=[humidity,pressure] target-unit=degC feature-units=[percent,hPa]

Actual native command answer

equation
"temperature = 13.705034643885003 + (-0.083280891215193192) * [humidity] + (0.025477293465290859) * [pressure]"

centeredEquation
"temperature = 34.5475 + (-0.083280891215193192) * ([humidity] - (55.500000000000007)) + (0.025477293465290859) * ([pressure] - (999.50000000000011))"

equationConvention
"target = intercept + sum(coefficients[i].coefficient * features[i]); fitted rows are evaluated in centered/scaled coordinates to reduce cancellation."

target
{"name":"temperature","columnIndex":3,"unit":"degC"}

features
[{"name":"humidity","columnIndex":1,"unit":"percent"},{"name":"pressure","columnIndex":2,"unit":"hPa"}]

intercept
13.705034643885003

coefficients
[{"feature":"humidity","coefficient":-0.083280891215193192,"unit":"degC/(percent)"},{"feature":"pressure","coefficient":0.025477293465290859,"unit":"degC/(hPa)"}]

metrics
{"rSquared":0.94955634720568838,"rmse":0.2347757958508522,"mae":0.2074963150136894,"residualSumSquares":1.3228721836176247,"residualSumSquaresRepresentable":true,"targetMean":34.5475,"constantTarget":false,"trainingOnly":true,"holdoutRows":0}

rowCount `26`

usedRows `24`

excludedRows `2`

rank `3`

parameterCount `3`

qrDiagonalRatio `1.4649456776352239`

residualDegreesOfFreedom `21`

source
{"id":"67b43c4e300b4882af7de00409533dc0","filename":"weather-example.csv","sha256":"7a64610e0a077d880955b264aec5c3616b348ab6a664368fe5a5b7d247c8905e","bytes":703}

warnings	`["Coefficients describe an empirical linear relationship in the supplied rows. They do not establish physical causality or validated future forecasts.","R-squared, RMSE and MAE use the same rows used to fit the model; no independent holdout validation was performed.","Rows with missing, nonnumeric, nonfinite or out-of-range selected cells were excluded; see every row index and reason."]`

Values above come from this command's returned result. A returned zero or false is data. Model verification below checks a separate procedural snapshot.

Original native product version	`3.1.1-candidate`
Recorded UTC	`2026-10-10T19:13:27.1560832Z`
Execution completion UTC	`2026-10-10T19:13:27.1560832Z`
Completion scope	`Synchronous command returned at this observed UTC; this is not independent numerical verification.`
Command elapsed (ms)	`59.1141`
Saved procedural descriptor SHA-256	`8851eb9b8dd83093c90097cf4c958108905e5bd1624bd3b1ad5c8a84cacc4117`
Field seed	`e1a83f29c3d5835b5b7f46bcf5334998f005abebc420be459e764fbeef6ae468`
Measurement seed	`1efd6137fe2bd4ff0be90691ad39eb2ec4dabcc886f34c2b08ef0283c9fd9949`
Record provenance	`Verified native archive snapshot: command c6324fef51f4453daac592afacd9b99c; stage completed.`
Selected native archive SHA-256	`27e53fc2dc36e1ff30182e965e0a5525350d305ee6d9eb4eca5f772e38620499`
Selected native archive stage	`accepted`
Response data scope	`Recorded native payload; chart scope is labeled per figure`

Empirical CSV equation and fitted data

Fitted equation	`temperature = 13.705034643885003 + (-0.083280891215193192) * [humidity] + (0.025477293465290859) * [pressure]`
Centered equation	`temperature = 34.5475 + (-0.083280891215193192) * ([humidity] - (55.500000000000007)) + (0.025477293465290859) * ([pressure] - (999.50000000000011))`
Equation convention	`target = intercept + sum(coefficients[i].coefficient * features[i]); fitted rows are evaluated in centered/scaled coordinates to reduce cancellation.`
Native fitting method	`centered/scaled column-pivoted Householder QR`
Model scope	`Bounded classical empirical CSV regression using actual supplied numeric rows. In-sample diagnostics only; no physical-law discovery, causal proof, quantum computation, holdout validation or future prediction is claimed.`

Equation representations from recorded data

Empirical in-sample fitted equation. Symbols map only to the actual returned CSV coefficients and column names; no circuit is inferred from measurement data.

Editable equation text

```
y_hat = 13.705034643885003 + (-0.083280891215193192) * x1 + (0.025477293465290859) * x2  y_hat is the fitted target: temperature; b0 = 13.705034643885003  x1 = CSV column "humidity"; b1 = -0.083280891215193192; unit = degC/(percent)  x2 = CSV column "pressure"; b2 = 0.025477293465290859; unit = degC/(hPa)  Recorded input-column equation: temperature = 13.705034643885003 + (-0.083280891215193192) * [humidity] + (0.025477293465290859) * [pressure]  Stable centered form: temperature = 34.5475 + (-0.083280891215193192) * ([humidity] - (55.500000000000007)) + (0.025477293465290859) * ([pressure] - (999.50000000000011))
```

Copy this text to an editable local draft. Editing a draft does not change the original CSV, native circuit or archived computation. The PDF text is selectable; the visual mathematics below is a presentation of the same symbols.

Conventional mathematical form

$$\hat{y} = b_0 + \sum_{j=1}^{2} b_j x_j$$

Symbol	Native coefficient	Native CSV column mapping	Unit
b0	13.705034643885003	Intercept	Target unit
b1	-0.083280891215193192	x1 = humidity	degC/(percent)
b2	0.025477293465290859	x2 = pressure	degC/(hPa)

The hat denotes a fitted estimate. p is the number of explicitly selected features. Each indexed variable maps to the native returned column and coefficient above; empirical fit data are not a declared circuit.

Copyable conventional notation (LaTeX) `\widehat{y}=b_0+\sum_{j=1}^{p}b_jx_j`

This equation fits the recorded numeric CSV rows. It is an empirical relationship within these data, not a physical law, a causal claim or a validated prediction. Export uses the completed fit and performs no fitting, sampling or quantum calculation.

Target column / recorded unit	temperature / degC
Original CSV filename	weather-example.csv
Original CSV SHA-256	7a64610e0a077d880955b264aec5c3616b348ab6a664368fe5a5b7d247c8905e
Original CSV bytes	703
rowCount	26
usedRows	24
excludedRows	2
rank	3
parameterCount	3
qrDiagonalRatio	1.4649456776352239
residualDegreesOfFreedom	21
Intercept	13.705034643885003

Feature	Recorded coefficient	Coefficient unit
humidity	-0.083280891215193192	degC/(percent)
pressure	0.025477293465290859	degC/(hPa)

Training fit metrics

rSquared	0.94955634720568838
rmse	0.2347757958508522
mae	0.2074963150136894
residualSumSquares	1.3228721836176247
residualSumSquaresRepresentable	True
targetMean	34.5475
constantTarget	False
trainingOnly	True
holdoutRows	0

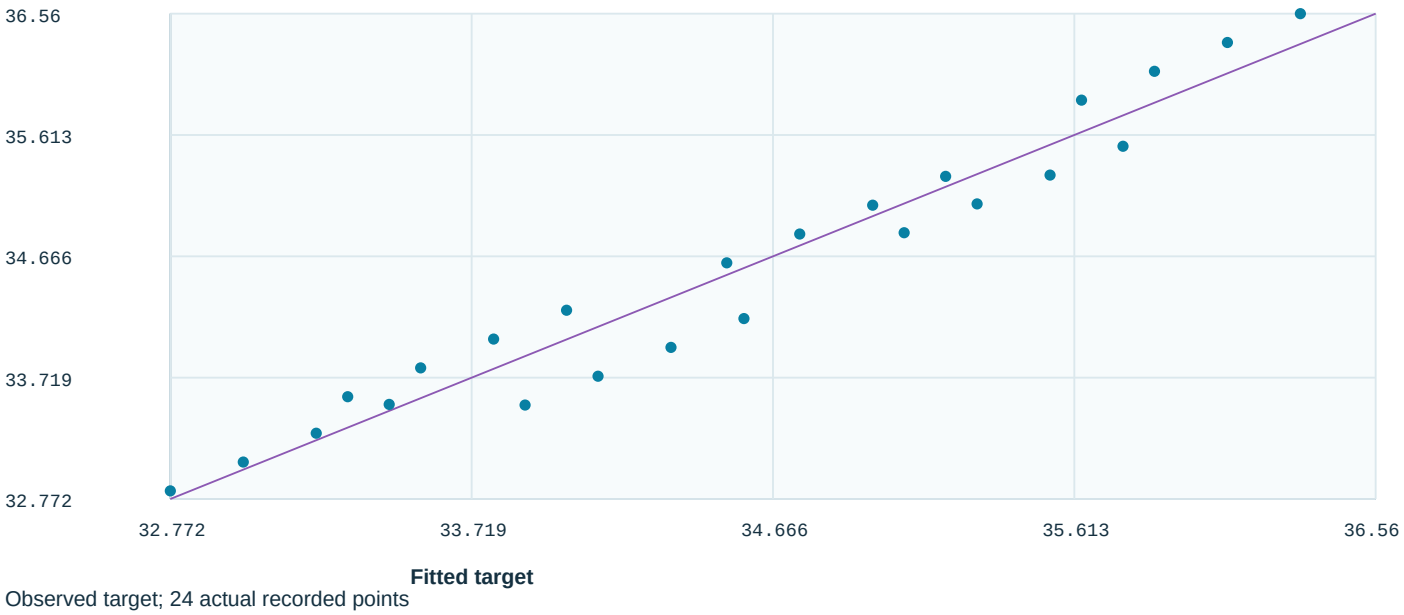
R-squared, RMSE and MAE describe the rows used to fit this equation. Lower residuals on these rows do not establish accuracy on unseen data. A null R-squared is retained as undefined; it is not replaced with zero.

Used column	Minimum	Maximum	Mean
Target: temperature	32.835	36.56	34.5475
humidity	38	73	55.500000000000007
pressure	985	1014	999.50000000000011

Recorded observations and residual figures

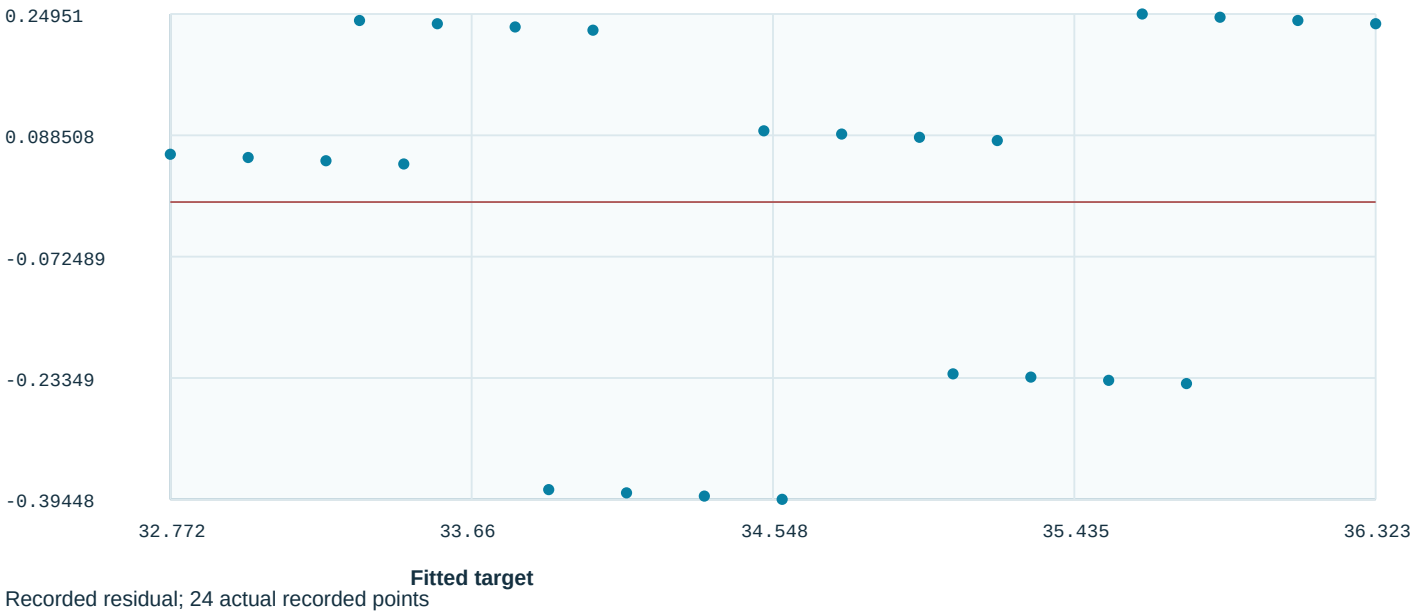
Recorded fitted data rows	24
Actual rows shown per figure	24

Observed values versus fitted values



Observed target; 24 actual recorded points

Residuals versus fitted values



Recorded residual; 24 actual recorded points

The diagonal is the equality reference, not fitted or additional data. The residual figure uses the native returned residuals; the horizontal zero reference is not an observation.

Excluded input rows

Recorded source row index	Exclusion reason
24	`feature humidity is missing, nonnumeric, nonfinite or outside +/-1e150`
25	`target temperature is missing, nonnumeric, nonfinite or outside +/-1e150`

Native fit warning: Coefficients describe an empirical linear relationship in the supplied rows. They do not establish physical causality or validated future forecasts.

Native fit warning: R-squared, RMSE and MAE use the same rows used to fit the model; no independent holdout validation was performed.

Native fit warning: Rows with missing, nonnumeric, nonfinite or out-of-range selected cells were excluded; see every row index and reason.

Native interpretation and evidence boundaries

Completion meaning	`A synchronous return or recorded final snapshot is shown. Command acceptance alone is not independent verification.`
Documented method for the requested operation	`Empirical ordinary least-squares fit of the selected CSV target column against the selected numeric feature columns with an intercept, using centered/scaled column-pivoted Householder QR. Rows and exclusions are returned explicitly. Training fit metrics do not establish causality or validated forecast accuracy.`

Native method / convention	`method=centered/scaled column-pivoted Householder QR; equation=temperature = 13.705034643885003 + (-0.083280891215193192) * [humidity] + (0.025477293465290859) * [pressure]; scope=Bounded classical empirical CSV regression using actual supplied numeric rows. In-sample diagnostics only; no physical-law discovery, causal proof, quantum computation, holdout validation or future prediction is claimed.`
Native echoed operands	`not returned`
Complex values	`[real, imaginary] stores both components of a complex number. An exactly returned [0,0] means zero amplitude/overlap; its phase is undefined. Probabilities are squared magnitudes, while density/operator entries are not individual probabilities.`
Returned native checks	`not returned`
Validation boundary	`Returned booleans/residuals describe native checks at the returned tolerance and scope. ok=true is a command return/acceptance flag. Independent verification is not established unless a matched native verification result and its coverage are explicitly returned; the exporter does not recompute or certify it.`
Command-local work returned	`physicalQuantumOperations=0`
Elapsed time scope	`Native response elapsedMs=59.1141; result elapsedMs=not returned; result totalMs=not returned. These are returned command/job timing scopes, not export time or a measured operation rate.`

Procedural model verification receipt

Receipt sampling	`NATIVE SAMPLE VERIFICATION NOT REQUESTED`
Verification phase	`not-requested`
Receipt release	`Archived native return available; no sample verification inferred`

The verification materializes sampled current-model predictions (amplitudes, spin and Born probabilities) and checks CPU / available GPU agreement. These are snapshot predictions, not measurement observations. Its PASS applies to this descriptor and sampled addresses; it does not independently certify arbitrary calculator outputs or every model address.

Folder receipt verification is not required for this command; inspect its own returned verification result if present.

The command result and returned checks are shown above. Complete accepted/final native fields, lifetime counters, hashes and verification rows remain in the matched TXT/JSON or Complete PDF view.

Recorded accounting

Scope: saved native status accompanying this recorded return. These values are not this command’s work delta and are not guaranteed to describe the current live session. An acceptance snapshot can precede later job or per-file verification evidence; those results retain their own job-local counts.

A recorded 0 is a genuine zero in this snapshot. Not returned means the field was absent; missing work is never replaced with 0. Defining a circuit does not add execution counts. Represented credits, actual procedural calculations, circuit execution and selected-register work are separate quantities, not FLOPS or physical quantum throughput.

Inactive counter groups are compacted below when every returned value in that scope is exactly 0 or false. Mathematical results, nonzero values and unknown fields are not changed. Complete PDF, TXT and JSON retain the exact individual counter rows.

Procedural domain (descriptor, not allocation)

Counter	Recorded value	Meaning
gridM	20000000	Address-grid width in the saved procedural descriptor.
gridL	25000000	Address-grid height in the saved procedural descriptor.
stateCount	500000000000000	Number of addressable single-qubit family states; not allocated amplitudes, calculated states or jointly entangled qubits.
hasPrepared	False	Whether procedural preparation is committed in this saved model; independent of a circuit definition.
migrationT	0	Saved procedural preparation parameter; not the circuit’s input state or run time.

Represented procedural credits

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Procedural model lifetime actual work

Inactive snapshot: all 7 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Selected tranche snapshot

Inactive snapshot: all 6 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Tranche lifetime actual work

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Circuit lifetime actual work

Inactive snapshot: all 11 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

q.input fit 8b288081317745c1bc941cc8c4daec1c target=success_probability features=[control_amplitude,pulse_ns] target-unit=probability feature-units=[relative,ns]

Actual native command answer

equation	"success_probability = 0.91 + (0.020000000000000028) * [control_amplitude] + (-0.004) * [pulse_ns]"
centeredEquation	"success_probability = 0.796 + (0.020000000000000028) * ([control_amplitude] - (0.9)) + (-0.004) * ([pulse_ns] - (33))"
equationConvention	"target = intercept + sum(coefficients[i].coefficient * features[i]); fitted rows are evaluated in centered/scaled coordinates to reduce cancellation."
target	{"name":"success_probability","columnIndex":2,"unit":"probability"}
features	[{"name":"control_amplitude","columnIndex":0,"unit":"relative"},{"name":"pulse_ns","columnIndex":1,"unit":"ns"}]
intercept	0.91
coefficients	[{"feature":"control_amplitude","coefficient":0.020000000000000028,"unit":"probability/(relative)"},{"feature":"pulse_ns","coefficient":-0.004,"unit":"probability/(ns)"}]
metrics	{"rSquared":0.99639435458947145,"rmse":0.0021602468994693083,"mae":0.0020000000000000157,"residualSumSquares":0.00011200000000000222,"residualSumSquaresRepresentable":true,"targetMean":0.79600000000000015,"constantTarget":false,"trainingOnly":true,"holdoutRows":0}
rowCount	24
usedRows	24
excludedRows	0
rank	3
parameterCount	3
qrDiagonalRatio	1.4652346924794752
residualDegreesOfFreedom	21
source	{"id":"8b288081317745c1bc941cc8c4daec1c","filename":"gate-measurements.csv","sha256":"cc603b81de12170ab62a520e74fa70d4068c43b3fe764a60d6004661c544144a","bytes":415}
warnings	["Coefficients describe an empirical linear relationship in the supplied rows. They do not establish physical causality or validated future forecasts.","R-squared, RMSE and MAE use the same rows used to fit the model; no independent holdout validation was performed."]

Values above come from this command's returned result. A returned zero or false is data. Model verification below checks a separate procedural snapshot.

Original native product version	3.1.1-candidate
Recorded UTC	2026-10-10T19:13:27.4227304Z
Execution completion UTC	2026-10-10T19:13:27.4227304Z
Completion scope	Synchronous command returned at this observed UTC; this is not independent numerical verification.
Command elapsed (ms)	1.3312000000000002
Saved procedural descriptor SHA-256	8851eb9b8dd83093c90097cf4c958108905e5bd1624bd3b1ad5c8a84cacc4117

Field seed	e1a83f29c3d5835b5b7f46bcf5334998f005abebc420be459e764fbeef6ae468
Measurement seed	1efd6137fe2bd4ff0be90691ad39eb2ec4dabcc886f34c2b08ef0283c9fd9949
Record provenance	Verified native archive snapshot: command 391b81144d02490d8b620810ad6929b2; stage completed.
Selected native archive SHA-256	f925f708777ddafc2fb92dfc14289662c5bb704507f7d31db06d8a3dc8c7f492
Selected native archive stage	accepted
Response data scope	Recorded native payload; chart scope is labeled per figure

Empirical CSV equation and fitted data

Fitted equation	success_probability = 0.91 + (0.020000000000000028) * [control_amplitude] + (-0.004) * [pulse_ns]
Centered equation	success_probability = 0.796 + (0.020000000000000028) * ([control_amplitude] - (0.9)) + (-0.004) * ([pulse_ns] - (33))
Equation convention	target = intercept + sum(coefficients[i].coefficient * features[i]); fitted rows are evaluated in centered/scaled coordinates to reduce cancellation.
Native fitting method	centered/scaled column-pivoted Householder QR
Model scope	Bounded classical empirical CSV regression using actual supplied numeric rows. In-sample diagnostics only; no physical-law discovery, causal proof, quantum computation, holdout validation or future prediction is claimed.

Equation representations from recorded data

Empirical in-sample fitted equation. Symbols map only to the actual returned CSV coefficients and column names; no circuit is inferred from measurement data.

Editable equation text

```
y_hat = 0.91 + (0.020000000000000028) * x1 + (-0.004) * x2  y_hat is the fitted target: success_probability; b0 = 0.91  x1 = CSV column "control_amplitude"; b1 = 0.020000000000000028; unit = probability/(relative)  x2 = CSV column "pulse_ns"; b2 = -0.004; unit = probability/(ns)  Recorded input-column equation: success_probability = 0.91 + (0.020000000000000028) * [control_amplitude] + (-0.004) * [pulse_ns]  Stable centered form: success_probability = 0.796 + (0.020000000000000028) * ([control_amplitude] - (0.9)) + (-0.004) * ([pulse_ns] - (33))
```

Copy this text to an editable local draft. Editing a draft does not change the original CSV, native circuit or archived computation. The PDF text is selectable; the visual mathematics below is a presentation of the same symbols.

Conventional mathematical form

$$\hat{y} = b_0 + \sum_{j=1}^{2} b_j x_j$$

Symbol	Native coefficient	Native CSV column mapping	Unit
b0	0.91	Intercept	Target unit

Symbol	Native coefficient	Native CSV column mapping	Unit
b1	0.020000000000000028	x1 = control_amplitude	probability/(relative)
b2	-0.004	x2 = pulse_ns	probability/(ns)

The hat denotes a fitted estimate. p is the number of explicitly selected features. Each indexed variable maps to the native returned column and coefficient above; empirical fit data are not a declared circuit.

Copyable conventional notation (LaTeX) `\widehat{y}=b_0+\sum_{j=1}^{p}b_jx_j`

This equation fits the recorded numeric CSV rows. It is an empirical relationship within these data, not a physical law, a causal claim or a validated prediction. Export uses the completed fit and performs no fitting, sampling or quantum calculation.

Target column / recorded unit `success_probability / probability`
Original CSV filename `gate-measurements.csv`
Original CSV SHA-256 `cc603b81de12170ab62a520e74fa70d4068c43b3fe764a60d6004661c544144a`
Original CSV bytes `415`
rowCount `24`
usedRows `24`
excludedRows `0`
rank `3`
parameterCount `3`
qrDiagonalRatio `1.4652346924794752`
residualDegreesOfFreedom `21`
Intercept `0.91`

Feature	Recorded coefficient	Coefficient unit
control_amplitude	0.020000000000000028	probability/(relative)
pulse_ns	-0.004	probability/(ns)

Training fit metrics

rSquared `0.99639435458947145`
rmse `0.0021602468994693083`
mae `0.0020000000000000157`
residualSumSquares `0.00011200000000000222`
residualSumSquaresRepresentable `True`
targetMean `0.79600000000000015`
constantTarget `False`
trainingOnly `True`
holdoutRows `0`

R-squared, RMSE and MAE describe the rows used to fit this equation. Lower residuals on these rows do not establish accuracy on unseen data. A null R-squared is retained as undefined; it is not replaced with zero.

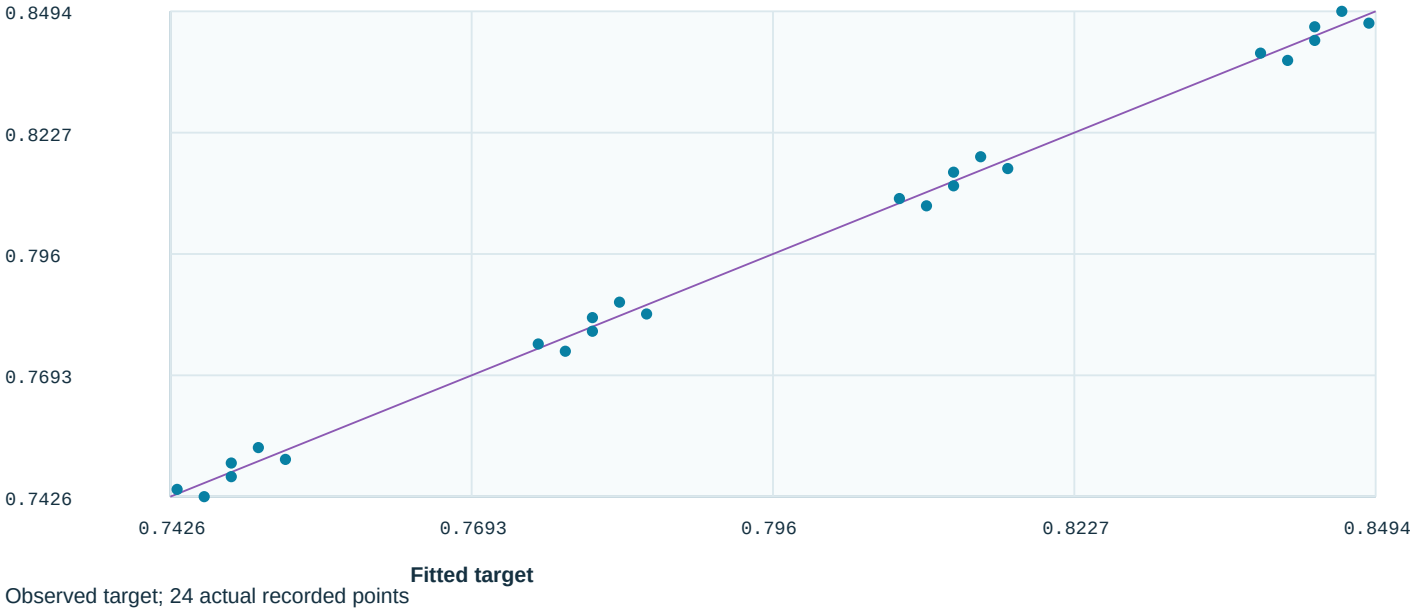
Used column	Minimum	Maximum	Mean
Target: success_probability	0.7426	0.8494	0.79600000000000015
control_amplitude	0.7	1.1	0.9
pulse_ns	20	46	33

Recorded observations and residual figures

Recorded fitted data rows 24

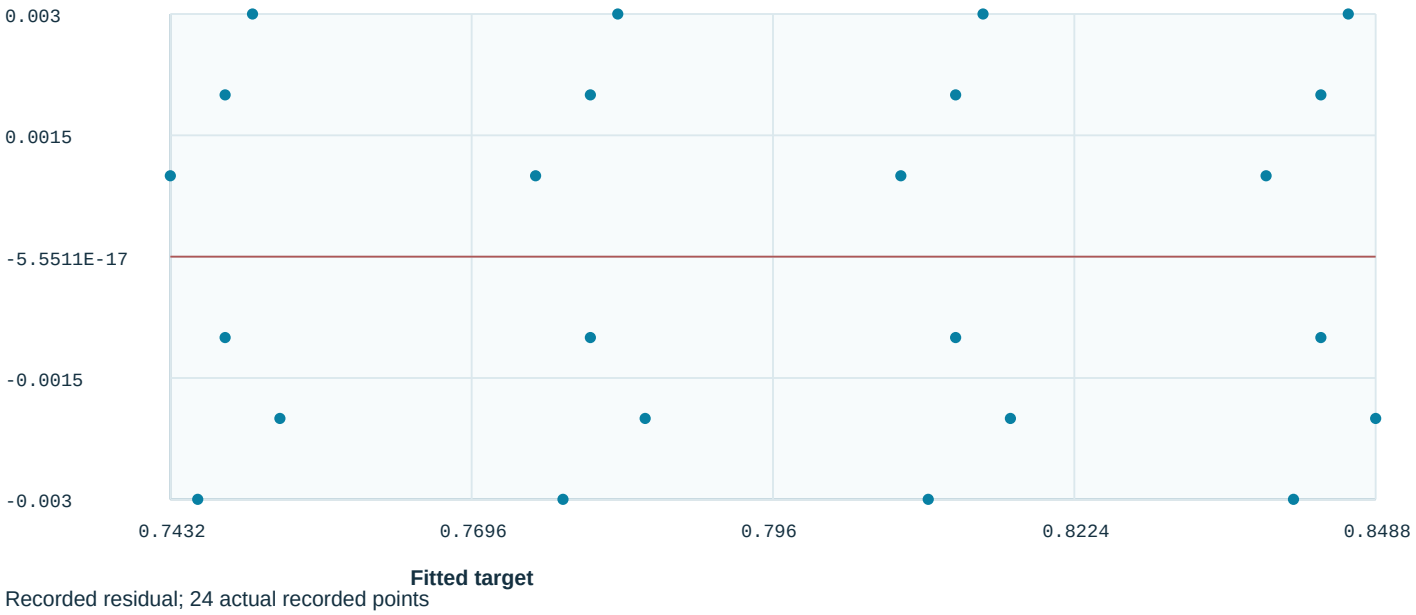
Actual rows shown per figure 24

Observed values versus fitted values



Observed target; 24 actual recorded points

Residuals versus fitted values



Recorded residual; 24 actual recorded points

The diagonal is the equality reference, not fitted or additional data. The residual figure uses the native returned residuals; the horizontal zero reference is not an observation.

Native fit warning: Coefficients describe an empirical linear relationship in the supplied rows. They do not establish physical causality or validated future forecasts.

Native fit warning: R-squared, RMSE and MAE use the same rows used to fit the model; no independent holdout validation was performed.

Native interpretation and evidence boundaries

Completion meaning	A synchronous return or recorded final snapshot is shown. Command acceptance alone is not independent verification.
Documented method for the requested operation	Empirical ordinary least-squares fit of the selected CSV target column against the selected numeric feature columns with an intercept, using centered/scaled column-pivoted Householder QR. Rows and exclusions are returned explicitly. Training fit metrics do not establish causality or validated forecast accuracy.
Native method / convention	method=centered/scaled column-pivoted Householder QR; equation=success_probability = 0.91 + (0.020000000000000028) * [control_amplitude] + (-0.004) * [pulse_ns]; scope=Bounded classical empirical CSV regression using actual supplied numeric rows. In-sample diagnostics only; no physical-law discovery, causal proof, quantum computation, holdout validation or future prediction is claimed.
Native echoed operands	not returned
Complex values	[real, imaginary] stores both components of a complex number. An exactly returned [0,0] means zero amplitude/overlap; its phase is undefined. Probabilities are squared magnitudes, while density/operator entries are not individual probabilities.
Returned native checks	not returned

Validation boundary	Returned booleans/residuals describe native checks at the returned tolerance and scope. ok=true is a command return/acceptance flag. Independent verification is not established unless a matched native verification result and its coverage are explicitly returned; the exporter does not recompute or certify it.
Command-local work returned	physicalQuantumOperations=0
Elapsed time scope	Native response elapsedMs=1.3312000000000002; result elapsedMs=not returned; result totalMs=not returned. These are returned command/job timing scopes, not export time or a measured operation rate.

Procedural model verification receipt

Receipt sampling	NATIVE SAMPLE VERIFICATION NOT REQUESTED
Verification phase	not-requested
Receipt release	Archived native return available; no sample verification inferred

The verification materializes sampled current-model predictions (amplitudes, spin and Born probabilities) and checks CPU / available GPU agreement. These are snapshot predictions, not measurement observations. Its PASS applies to this descriptor and sampled addresses; it does not independently certify arbitrary calculator outputs or every model address.

Folder receipt verification is not required for this command; inspect its own returned verification result if present.

The command result and returned checks are shown above. Complete accepted/final native fields, lifetime counters, hashes and verification rows remain in the matched TXT/JSON or Complete PDF view.

Recorded accounting

Scope: saved native status accompanying this recorded return. These values are not this command's work delta and are not guaranteed to describe the current live session. An acceptance snapshot can precede later job or per-file verification evidence; those results retain their own job-local counts.

A recorded 0 is a genuine zero in this snapshot. Not returned means the field was absent; missing work is never replaced with 0. Defining a circuit does not add execution counts. Represented credits, actual procedural calculations, circuit execution and selected-register work are separate quantities, not FLOPS or physical quantum throughput.

Inactive counter groups are compacted below when every returned value in that scope is exactly 0 or false. Mathematical results, nonzero values and unknown fields are not changed. Complete PDF, TXT and JSON retain the exact individual counter rows.

Procedural domain (descriptor, not allocation)

Counter	Recorded value	Meaning
gridM	20000000	Address-grid width in the saved procedural descriptor.
gridL	25000000	Address-grid height in the saved procedural descriptor.
stateCount	500000000000000	Number of addressable single-qubit family states; not allocated amplitudes, calculated states or jointly entangled qubits.

Counter	Recorded value	Meaning
hasPrepared	`False`	`Whether procedural preparation is committed in this saved model; independent of a circuit definition.`
migrationT	`0`	`Saved procedural preparation parameter; not the circuit’s input state or run time.`

Represented procedural credits

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Procedural model lifetime actual work

Inactive snapshot: all 7 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Selected tranche snapshot

Inactive snapshot: all 6 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Tranche lifetime actual work

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Circuit lifetime actual work

Inactive snapshot: all 11 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

q.input fit 4f79eb458d544d7188a14efb454c935a target=success_probability features=[depth,two_qubit_gates] target-unit=probability feature-units=[count,count]

Actual native command answer

equation	"success_probability = 0.9857950819672131 + (-0.011991803278688526) * [depth] + (-0.008191256830601094) * [two_qubit_gates]"
centeredEquation	"success_probability = 0.853 + (-0.011991803278688526) * ([depth] - (7.9999999999999991)) + (-0.008191256830601094) * ([two_qubit_gates] - (4.5000000000000009))"
equationConvention	"target = intercept + sum(coefficients[i].coefficient * features[i]); fitted rows are evaluated in centered/scaled coordinates to reduce cancellation."
target	{"name":"success_probability","columnIndex":2,"unit":"probability"}
features	[{"name":"depth","columnIndex":0,"unit":"count"},{"name":"two_qubit_gates","columnIndex":1,"unit":"count"}]
intercept	0.9857950819672131
coefficients	[{"feature":"depth","coefficient":-0.011991803278688526,"unit":"probability/(count)"},{"feature":"two_qubit_gates","coefficient":-0.008191256830601094,"unit":"probability/(count)"}]
metrics	{"rSquared":0.99790149585679944,"rmse":0.0021155164466544964,"mae":0.0019125683060109329,"residualSumSquares":0.00010740983606557596,"residualSumSquaresRepresentable":true,"targetMean":0.85300000000000009,"constantTarget":false,"trainingOnly":true,"holdoutRows":0}
rowCount	24
usedRows	24
excludedRows	0
rank	3
parameterCount	3
qrDiagonalRatio	1.5306522057160432
residualDegreesOfFreedom	21
source	{"id":"4f79eb458d544d7188a14efb454c935a","filename":"circuit-measurements.csv","sha256":"dd554387c0375775801cdcf1be23eb6950b689fba3b456a8aa35cc6427f3b8cb","bytes":311}
warnings	["Coefficients describe an empirical linear relationship in the supplied rows. They do not establish physical causality or validated future forecasts.","R-squared, RMSE and MAE use the same rows used to fit the model; no independent holdout validation was performed."]

Values above come from this command's returned result. A returned zero or false is data. Model verification below checks a separate procedural snapshot.

Original native product version	3.1.1-candidate
Recorded UTC	2026-10-10T19:13:27.8469816Z
Execution completion UTC	2026-10-10T19:13:27.8469816Z
Completion scope	Synchronous command returned at this observed UTC; this is not independent numerical verification.
Command elapsed (ms)	0.92480000000000007
Saved procedural descriptor SHA-256	8851eb9b8dd83093c90097cf4c958108905e5bd1624bd3b1ad5c8a84cacc4117
Field seed	e1a83f29c3d5835b5b7f46bcf5334998f005abebc420be459e764fbeef6ae468

Measurement seed	`1efd6137fe2bd4ff0be90691ad39eb2ec4dabcc886f34c2b08ef0283c9fd9949`
Record provenance	`Verified native archive snapshot: command e526e6688ea846f59dd22a3e7cf510ad; stage completed.`
Selected native archive SHA-256	`3384e3acfcb659903de7251b36843212ee10afa165b63213cf6aee6636f604e5`
Selected native archive stage	`accepted`
Response data scope	`Recorded native payload; chart scope is labeled per figure`

Empirical CSV equation and fitted data

Fitted equation	`success_probability = 0.9857950819672131 + (-0.011991803278688526) * [depth] + (-0.008191256830601094) * [two_qubit_gates]`
Centered equation	`success_probability = 0.853 + (-0.011991803278688526) * ([depth] - (7.9999999999999991)) + (-0.008191256830601094) * ([two_qubit_gates] - (4.5000000000000009))`
Equation convention	`target = intercept + sum(coefficients[i].coefficient * features[i]); fitted rows are evaluated in centered/scaled coordinates to reduce cancellation.`
Native fitting method	`centered/scaled column-pivoted Householder QR`
Model scope	`Bounded classical empirical CSV regression using actual supplied numeric rows. In-sample diagnostics only; no physical-law discovery, causal proof, quantum computation, holdout validation or future prediction is claimed.`

Equation representations from recorded data

Empirical in-sample fitted equation. Symbols map only to the actual returned CSV coefficients and column names; no circuit is inferred from measurement data.

Editable equation text

```
y_hat = 0.9857950819672131 + (-0.011991803278688526) * x1 + (-0.008191256830601094) * x2  y_hat is the fitted target: success_probability; b0 = 0.9857950819672131  x1 = CSV column "depth"; b1 = -0.011991803278688526; unit = probability/(count)  x2 = CSV column "two_qubit_gates"; b2 = -0.008191256830601094; unit = probability/(count)  Recorded input-column equation: success_probability = 0.9857950819672131 + (-0.011991803278688526) * [depth] + (-0.008191256830601094) * [two_qubit_gates]
Stable centered form: success_probability = 0.853 + (-0.011991803278688526) * ([depth] - (7.9999999999999991)) + (-0.008191256830601094) * ([two_qubit_gates] - (4.5000000000000009))
```

Copy this text to an editable local draft. Editing a draft does not change the original CSV, native circuit or archived computation. The PDF text is selectable; the visual mathematics below is a presentation of the same symbols.

Conventional mathematical form

$$\hat{y} = b_0 + \sum_{j=1}^{2} b_j x_j$$

Symbol	Native coefficient	Native CSV column mapping	Unit
b0	`0.9857950819672131`	`Intercept`	`Target unit`

Symbol	Native coefficient	Native CSV column mapping	Unit
b1	-0.011991803278688526	x1 = depth	probability/(count)
b2	-0.008191256830601094	x2 = two_qubit_gates	probability/(count)

The hat denotes a fitted estimate. p is the number of explicitly selected features. Each indexed variable maps to the native returned column and coefficient above; empirical fit data are not a declared circuit.

Copyable conventional notation (LaTeX) `\widehat{y}=b_0+\sum_{j=1}^{p}b_jx_j`

This equation fits the recorded numeric CSV rows. It is an empirical relationship within these data, not a physical law, a causal claim or a validated prediction. Export uses the completed fit and performs no fitting, sampling or quantum calculation.

Target column / recorded unit `success_probability / probability`
Original CSV filename `circuit-measurements.csv`
Original CSV SHA-256 `dd554387c0375775801cdcf1be23eb6950b689fba3b456a8aa35cc6427f3b8cb`
Original CSV bytes `311`
rowCount `24`
usedRows `24`
excludedRows `0`
rank `3`
parameterCount `3`
qrDiagonalRatio `1.5306522057160432`
residualDegreesOfFreedom `21`
Intercept `0.9857950819672131`

Feature	Recorded coefficient	Coefficient unit
depth	-0.011991803278688526	probability/(count)
two_qubit_gates	-0.008191256830601094	probability/(count)

Training fit metrics

rSquared `0.99790149585679944`
rmse `0.0021155164466544964`
mae `0.0019125683060109329`
residualSumSquares `0.00010740983606557596`
residualSumSquaresRepresentable `True`
targetMean `0.85300000000000009`
constantTarget `False`
trainingOnly `True`
holdoutRows `0`

R-squared, RMSE and MAE describe the rows used to fit this equation. Lower residuals on these rows do not establish accuracy on unseen data. A null R-squared is retained as undefined; it is not replaced with zero.

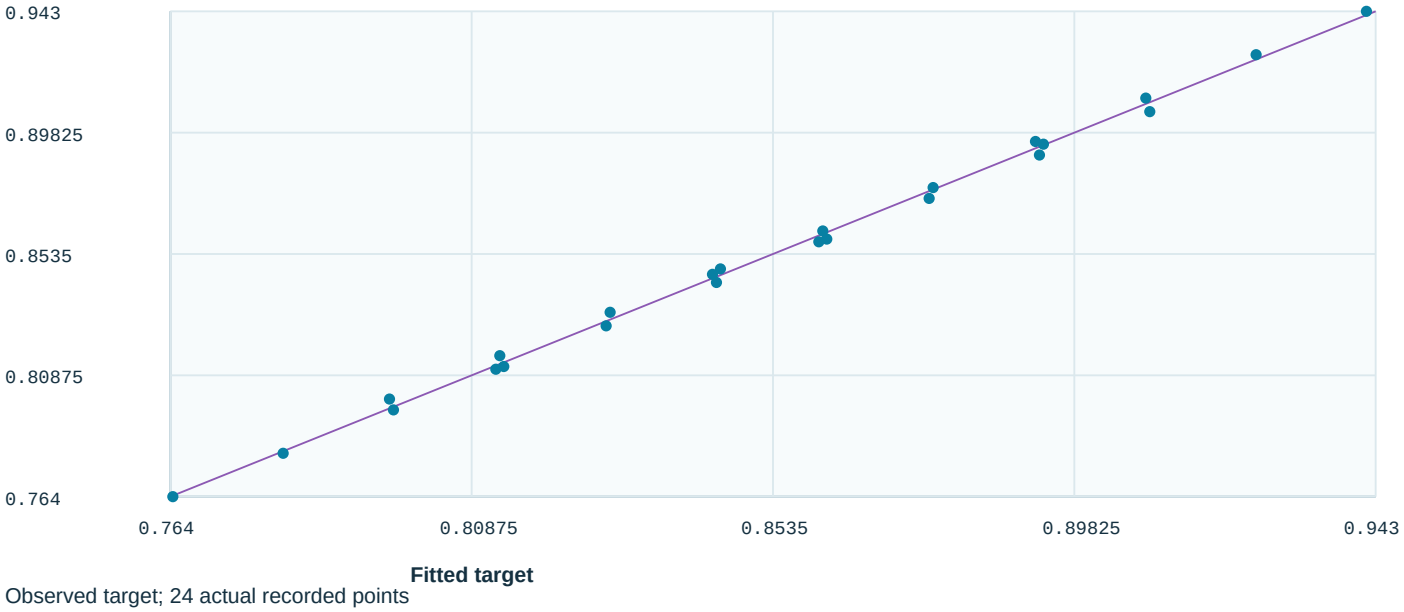
Used column	Minimum	Maximum	Mean
Target: success_probability	0.764	0.943	0.85300000000000009
depth	3	13	7.9999999999999991
two_qubit_gates	1	8	4.5000000000000009

Recorded observations and residual figures

Recorded fitted data rows 24

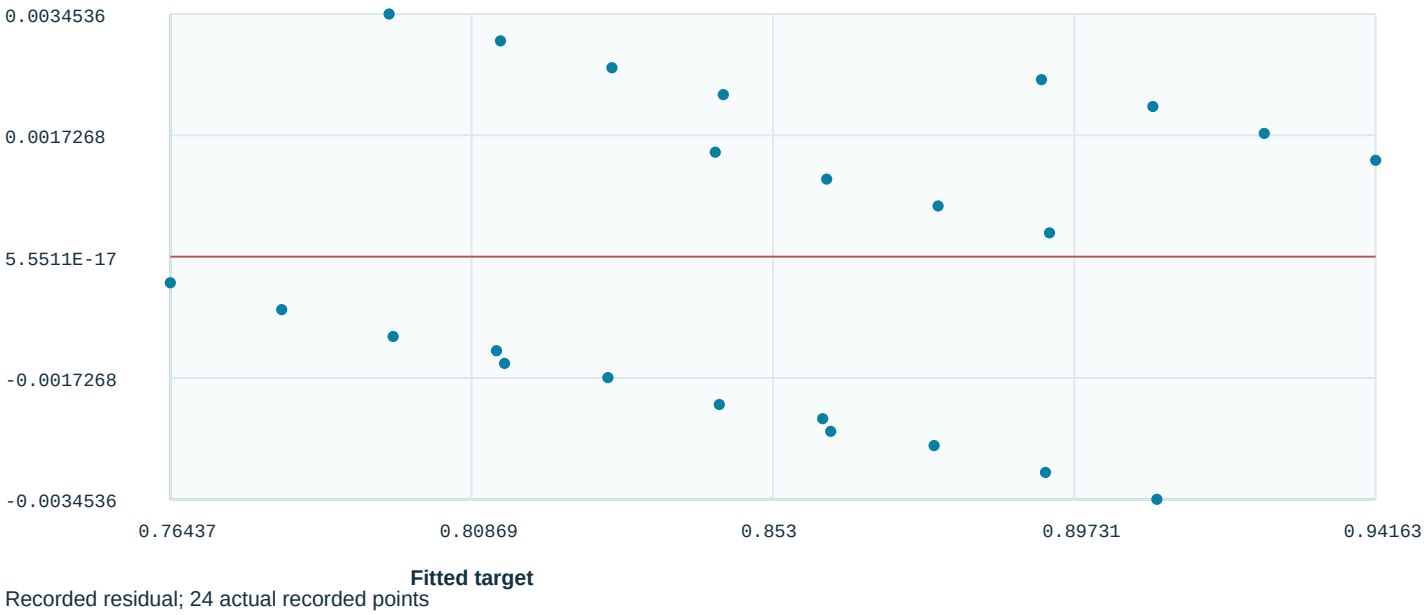
Actual rows shown per figure 24

Observed values versus fitted values



Observed target; 24 actual recorded points

Residuals versus fitted values



Recorded residual; 24 actual recorded points

The diagonal is the equality reference, not fitted or additional data. The residual figure uses the native returned residuals; the horizontal zero reference is not an observation.

Native fit warning: Coefficients describe an empirical linear relationship in the supplied rows. They do not establish physical causality or validated future forecasts.

Native fit warning: R-squared, RMSE and MAE use the same rows used to fit the model; no independent holdout validation was performed.

Native interpretation and evidence boundaries

Completion meaning	A synchronous return or recorded final snapshot is shown. Command acceptance alone is not independent verification.
Documented method for the requested operation	Empirical ordinary least-squares fit of the selected CSV target column against the selected numeric feature columns with an intercept, using centered/scaled column-pivoted Householder QR. Rows and exclusions are returned explicitly. Training fit metrics do not establish causality or validated forecast accuracy.
Native method / convention	method=centered/scaled column-pivoted Householder QR; equation=success_probability = 0.9857950819672131 + (-0.011991803278688526) * [depth] + (-0.008191256830601094) * [two_qubit_gates]; scope=Bounded classical empirical CSV regression using actual supplied numeric rows. In-sample diagnostics only; no physical-law discovery, causal proof, quantum computation, holdout validation or future prediction is claimed.
Native echoed operands	not returned
Complex values	[real, imaginary] stores both components of a complex number. An exactly returned [0,0] means zero amplitude/overlap; its phase is undefined. Probabilities are squared magnitudes, while density/operator entries are not individual probabilities.
Returned native checks	not returned

Validation boundary	Returned booleans/residuals describe native checks at the returned tolerance and scope. ok=true is a command return/acceptance flag. Independent verification is not established unless a matched native verification result and its coverage are explicitly returned; the exporter does not recompute or certify it.
Command-local work returned	physicalQuantumOperations=0
Elapsed time scope	Native response elapsedMs=0.92480000000000007; result elapsedMs=not returned; result totalMs=not returned. These are returned command/job timing scopes, not export time or a measured operation rate.

Procedural model verification receipt

Receipt sampling	NATIVE SAMPLE VERIFICATION NOT REQUESTED
Verification phase	not-requested
Receipt release	Archived native return available; no sample verification inferred

The verification materializes sampled current-model predictions (amplitudes, spin and Born probabilities) and checks CPU / available GPU agreement. These are snapshot predictions, not measurement observations. Its PASS applies to this descriptor and sampled addresses; it does not independently certify arbitrary calculator outputs or every model address.

Folder receipt verification is not required for this command; inspect its own returned verification result if present.

The command result and returned checks are shown above. Complete accepted/final native fields, lifetime counters, hashes and verification rows remain in the matched TXT/JSON or Complete PDF view.

Recorded accounting

Scope: saved native status accompanying this recorded return. These values are not this command's work delta and are not guaranteed to describe the current live session. An acceptance snapshot can precede later job or per-file verification evidence; those results retain their own job-local counts.

A recorded 0 is a genuine zero in this snapshot. Not returned means the field was absent; missing work is never replaced with 0. Defining a circuit does not add execution counts. Represented credits, actual procedural calculations, circuit execution and selected-register work are separate quantities, not FLOPS or physical quantum throughput.

Inactive counter groups are compacted below when every returned value in that scope is exactly 0 or false. Mathematical results, nonzero values and unknown fields are not changed. Complete PDF, TXT and JSON retain the exact individual counter rows.

Procedural domain (descriptor, not allocation)

Counter	Recorded value	Meaning
gridM	20000000	Address-grid width in the saved procedural descriptor.
gridL	25000000	Address-grid height in the saved procedural descriptor.
stateCount	500000000000000	Number of addressable single-qubit family states; not allocated amplitudes, calculated states or jointly entangled qubits.

Counter	Recorded value	Meaning
hasPrepared	`False`	`Whether procedural preparation is committed in this saved model; independent of a circuit definition.`
migrationT	`0`	`Saved procedural preparation parameter; not the circuit’s input state or run time.`

Represented procedural credits

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Procedural model lifetime actual work

Inactive snapshot: all 7 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Selected tranche snapshot

Inactive snapshot: all 6 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Tranche lifetime actual work

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Circuit lifetime actual work

Inactive snapshot: all 11 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

q.circuit run shots 8 basis z

Actual native command answer

id	"9f054e1d3ade4aadaaa97985e5c4dd75"
active	false
phase	"completed"
programName	"file_6eaa955cc2b541cc"
programHash	"059509d74130748904a3f46d60cb5a48486f47f347a7602684a5e363ad0f0863"
qubits	1
initialBasis	"0"
basis	"z"
requestedShots	"8"
completedShots	"8"
measuredQubits	"8"
counts	{"executedCircuitInstructions":"9","backendGateApplications":"9","actualAmplitudeWrites":"52","actualTensorProducts":"0","actualPreparedQubits":"9","completedShots":"8","measuredQubits":"8","uniformDraws":"8","exactCPReductions":"0","queryCount":"9","coherentOutputExecutions":"1"}
counterStart	"0"
counterEnd	"8"
error	""
checkpointError	""

Values above come from this command's returned result. A returned zero or false is data. Model verification below checks a separate procedural snapshot.

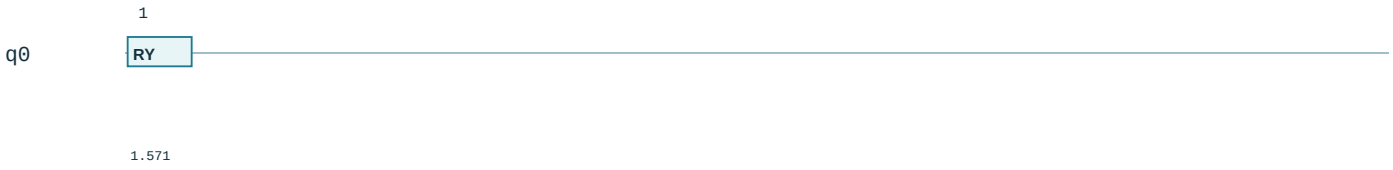
Original native product version	3.1.1-candidate
Recorded UTC	2026-10-10T19:13:29.7487102Z
Execution completion UTC	2026-10-10T19:13:30.2984004Z
Completion scope	Matched selected native sequence completion barrier observed at this UTC; this is execution evidence, not independent numerical verification.
Command elapsed (ms)	68.8963
Saved procedural descriptor SHA-256	8851eb9b8dd83093c90097cf4c958108905e5bd1624bd3b1ad5c8a84cacc4117
Field seed	e1a83f29c3d5835b5b7f46bcf5334998f005abebc420be459e764fbeef6ae468
Measurement seed	1efd6137fe2bd4ff0be90691ad39eb2ec4dabcc886f34c2b08ef0283c9fd9949
Record provenance	Verified native archive snapshot: sequence 97e97fcb9ebb42279c3169e873a0b48b; stage completed.
Original input filename	gate-rotated-qubit-8-shots.csv
Original input format	csv-gates
Original input SHA-256	8938d908c311e756f7a8094149c8ca03b8fbf4cd9a5cd3f9a5f455cd909314dd
Selected native archive SHA-256	5c28b7eca2c5748a8dfa82cf53bab26f37d16cc517732180adcdf1f376d1cc70
Selected native archive stage	completed
Response data scope	Recorded native payload; chart scope is labeled per figure
Job phase	completed
Job elapsed (ms)	267.53790000000004

No complete native TXT snapshot matches this recorded result. The histogram below uses this response's own rows and completed-shot count.

programName	`file_6eaa955cc2b541cc`
programHash	`059509d74130748904a3f46d60cb5a48486f47f347a7602684a5e363ad0f0863`
qubits	`1`
initialBasis	`0`
basis	`z`
phase	`completed`
requestedShots	`8`
completedShots	`8`
measuredQubits	`8`
elapsedMs	`267.53790000000004`
seed	`421d696fd15594faa8c7ed33687909240d8fe2f6e3aeccffa48705b305bcf05f`
counterStart	`0`
counterEnd	`8`
error	`Not returned`
checkpointError	`Not returned`

Circuit gate sequence

Qubits / input basis	`1 / 0`



Layers 1-1; full exact angles are retained in the program descriptor.

Wires are indexed q0 through q0. Controls are dots; targets are labeled boxes. Gate angles below each layer are in radians. Diagram layers follow program order.

Exact gate operands

Layer / gate	Qubit indices	Angles (radians)
1. ry	`[0]`	`["1.5707963267948966"]`

Equation representations from recorded data

Symbolic declared circuit evolution from actual retained chronological program operands. Local matrices are native-convention formulas with recorded angle bindings; no full-register matrix, numerical state, probability, amplitude or extra shot is calculated by export. Histogram and spin figures remain separately recorded actual results.

Editable equation text

```
|psi_out> = U[1] |0>  Computational-basis index b = 0; register width n = 1; |b> = |0>.  Gate rows are chronological; the first row acts first (rightmost in the operator product). Global bitstrings have qubit0 at the right. Local two-operand matrices use |operand1,operand0> = |00>,|01>,|10>,|11>: the first listed operand is the local least-significant bit, including a controlled gate’s control.  U[1] = RY; operands = [0]; all = False; angle radians = [1.5707963267948966]    Local symbolic operator: [["cos(a1/2)","-sin(a1/2)"],["sin(a1/2)","cos(a1/2)"]]    a1 = 1.5707963267948966 radians  Born: P(s | basis) = |<s;basis|psi_out>|^2 = |<s| B_basis |psi_out>|^2; B_basis is the measurement rotation (basis-ket matrix adjoint): B_Z = I, B_X = tensor(H), B_Y = tensor(H*S_dagger). These are mathematical basis equivalents, not extra executed gates. Recorded measurement basis: z  Local Bloch: rho_q = partial_trace_other_qubits(|psi_out><psi_out|); x=Tr(rho_q X)=2Re(rho_q[0,1]); y=Tr(rho_q Y)=-2Im(rho_q[0,1]); z=Tr(rho_q Z)=rho_q[0,0]-rho_q[1,1]. Entangled local states can have x=y=z=0.
```

Copy this text to an editable local draft. Editing a draft does not change the original CSV, native circuit or archived computation. The PDF text is selectable; the visual mathematics below is a presentation of the same symbols.

Conventional mathematical form

$$|\psi_{out}> = U_1 \quad |0>$$

$$P(s \mid B) = |<s| B | \quad \psi_{out} >|^2$$

$$r_{q,k} = Tr(\rho_q \quad \sigma_k)$$

Operators act from right to left; U1 is the first archived gate instruction when any gates are present. B is the mathematical measurement rotation, equivalent to projecting onto the selected basis kets; it does not add gates executed by this exporter. Local reduced density matrices allow zero Bloch vectors for entangled mixed marginal states.

RY - local mathematical operator

Basis / operand convention `|0>,|1>`

$$\begin{bmatrix} cos(a1/2) & -sin(a1/2) \\ sin(a1/2) & cos(a1/2) \end{bmatrix}$$

Copyable conventional notation (LaTeX)

```
|\psi_{\mathrm{out}}\rangle=U_1|b\rangle;\quad P(s\mid B)=|\langle s|B|\psi_{\mathrm{out}}\rangle|^2;\quad r_{q,k}=\mathrm{Tr}(\rho_q\sigma_k)
```

Histogram scope `Complete recorded outcome distribution`

Returned histogram shot mass `8 / 8`

Completed joint shots / requested



Observed joint outcomes - most frequent 24 shown



Bars show returned counts, not probability amplitudes. Exact rows follow; omitted preview rows are explicitly identified.

Circuit work accounting

Job-local actual work for this recorded circuit run, identified by its native job id and programHash. These counts remain separate from the saved circuit lifetime totals and from the procedural model's CPU/GPU verification counters.

executedCircuitInstructions	9
backendGateApplications	9
actualAmplitudeWrites	52
actualTensorProducts	0
actualPreparedQubits	9
completedShots	8
measuredQubits	8
uniformDraws	8
exactCPReductions	0
queryCount	9
coherentOutputExecutions	1

Actual work includes independent shot replays and the separately retained coherent-output execution. Exact classical-input controlled-phase reductions are recorded. These units are not physical quantum gate throughput or FLOPS.

Retained circuit state before measurement

programHash	059509d74130748904a3f46d60cb5a48486f47f347a7602684a5e363ad0f0863
capturedUtc	2026-10-10T19:13:30.0699711Z
premeasurement	True
selectedQubits	1
returnedQubits	1
omittedQubits	0
unrequestedQubits	0
qubitLimit	4
maximumFactorEntries	65536
scope	Actual reduced Pauli expectations for the first at most four qubits of the retained coherent circuit output before independent measurement shots. Each queried coherent factor has at most 65536 stored entries. Omitted qubits were not evaluated and are not assigned zero vectors. This capture is separate from the saved procedural model and from postmeasurement shot states.

verificationScope	`These are actual circuit-state calculations. The procedural-model q.verify CPU/GPU sampling gate does not independently verify these circuit snapshots.`

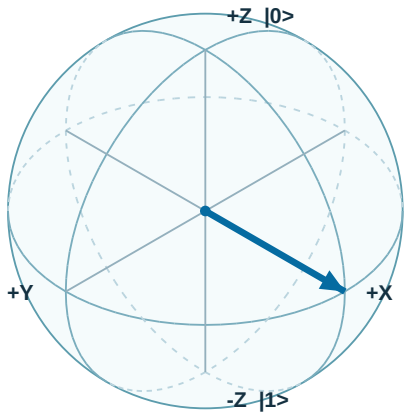
These local Pauli expectations were captured from this run’s actual coherent output before shot measurements. They are not reconstructed from the histogram, do not describe a later edited circuit, and are separate from procedural q.verify sampling.

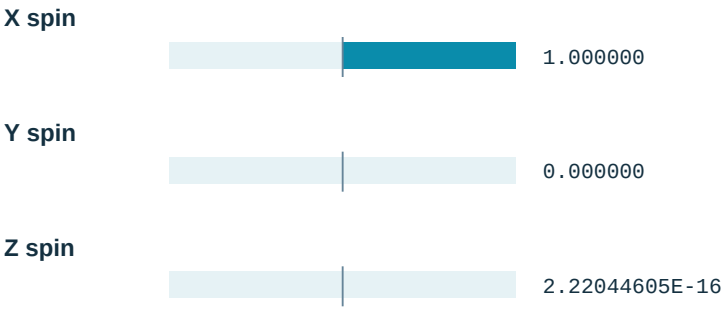
Circuit local spin / qubit 0

Spin figure source	Selected coherent circuit premeasurement output; reduced single-qubit state.
qubit	0
premeasurement	True
stateCharacter	pure-single-qubit
purity	1
blochLengthSquared	1
pure	True
mixed	False
exactlyZeroBloch	False
directionDefined	True
directionResolvedAtTolerance	True
directionTolerance	2E-09
spinScope	X/Y/Z are reduced Pauli expectations from the actual retained coherent register, not the separately saved procedural model or a postmeasurement shot. Entanglement can give a mixed local state. An exact zero vector has no preferred direction; a nonzero vector below tolerance retains its components and is labelled near-maximally mixed.

This is a retained coherent-state prediction, separate from observed shot counts and the saved procedural model checked by q.verify. A zero local vector has no preferred direction; |0> is the north-pole basis label, not a zero vector.

Bloch sphere and spin arrow





Vector length = 1.000000

Figure: orthographic view of the reported X/Y/Z expectations.
Arrow length is preserved in 3D; projection may look shorter.
Near-unit vector: close to the surface at displayed precision.
`Recorded XYZ = (1, 0, 2.22044604925031E-16)`

Returned histogram rows

Histogram rows included	2

Bitstring (qubit 0 at right)	Completed joint shots
0	3
1	5

Native interpretation and evidence boundaries

Completion meaning	A synchronous return or recorded final snapshot is shown. Command acceptance alone is not independent verification.
Documented method for the requested operation	Replay the selected circuit independently for each requested joint measurement shot using seeded Born sampling; a separate coherent-output execution is retained. Acceptance is not completion; matched job-local work and histogram fields describe actual execution.
Native method / convention	bitOrder=Most significant displayed bit at left; qubit 0 at right.; scope=Full joint bitstrings from independently replayed coherent registers and sequential collapsing Born measurements. Histograms contain completed shots only; partial shot work and consumed random draws are counted separately.
Native echoed operands	qubits=1; basis=z; initialBasis=0; requestedShots=8
Complex values	[real, imaginary] stores both components of a complex number. An exactly returned [0,0] means zero amplitude/overlap; its phase is undefined. Probabilities are squared magnitudes, while density/operator entries are not individual probabilities.
Returned native checks	not returned
Validation boundary	Returned booleans/residuals describe native checks at the returned tolerance and scope. ok=true is a command return/acceptance flag. Independent verification is not established unless a matched native verification result and its coverage are explicitly returned; the exporter does not recompute or certify it.
Command-local work returned	completedShots=8
Elapsed time scope	Native response elapsedMs=not returned; result elapsedMs=267.53790000000004; result totalMs=not returned. These are returned command/job timing scopes, not export time or a measured operation rate.

Procedural model verification receipt

Receipt sampling	NATIVE SAMPLE VERIFICATION NOT REQUESTED
Verification phase	not-requested
Receipt release	Archived native return available; no sample verification inferred

The verification materializes sampled current-model predictions (amplitudes, spin and Born probabilities) and checks CPU / available GPU agreement. These are snapshot predictions, not measurement observations. Its PASS applies to this descriptor and sampled addresses; it does not independently certify arbitrary calculator outputs or every model address.

Folder receipt verification is not required for this command; inspect its own returned verification result if present.

The command result and returned checks are shown above. Complete accepted/final native fields, lifetime counters, hashes and verification rows remain in the matched TXT/JSON or Complete PDF view.

Recorded accounting

Scope: saved native status accompanying this recorded return. These values are not this command's work delta and are not guaranteed to describe the current live session. An acceptance snapshot can precede later job or per-file verification evidence; those results retain their own job-local counts.

A recorded 0 is a genuine zero in this snapshot. Not returned means the field was absent; missing work is never replaced with 0. Defining a circuit does not add execution counts. Represented credits, actual procedural calculations, circuit execution and selected-register work are separate quantities, not FLOPS or physical quantum throughput.

Inactive counter groups are compacted below when every returned value in that scope is exactly 0 or false. Mathematical results, nonzero values and unknown fields are not changed. Complete PDF, TXT and JSON retain the exact individual counter rows.

Procedural domain (descriptor, not allocation)

Counter	Recorded value	Meaning
gridM	20000000	Address-grid width in the saved procedural descriptor.
gridL	25000000	Address-grid height in the saved procedural descriptor.
stateCount	500000000000000	Number of addressable single-qubit family states; not allocated amplitudes, calculated states or jointly entangled qubits.
hasPrepared	False	Whether procedural preparation is committed in this saved model; independent of a circuit definition.
migrationT	0	Saved procedural preparation parameter; not the circuit's input state or run time.

Represented procedural credits

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Procedural model lifetime actual work

Inactive snapshot: all 7 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Selected tranche snapshot

Inactive snapshot: all 6 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Tranche lifetime actual work

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Circuit lifetime actual work

Counter	Recorded value	Meaning
actualPreparedQubits	9	Qubits actually prepared for circuit shot replays and retained coherent-output executions.
executedCircuitInstructions	9	Stored program instructions actually traversed by circuit executions; not the definition’s gateCount.
backendGateApplications	9	Actual local/joint backend gate calls; separate from program instruction count and exact CP reductions.
actualAmplitudeWrites	52	Actual circuit backend amplitude writes; not physical gates or FLOPS.
actualTensorProducts	0	Actual circuit backend tensor-product arithmetic work.
completedShots	8	Completed joint circuit measurement shots across recorded jobs; not probabilities or requested shots.
measuredQubits	8	Completed circuit qubit measurements, including work from partial shots.
uniformDraws	8	Seeded random variates actually consumed by circuit measurements; a cancelled partial draw can still count.
exactCPReductions	0	Recorded exact controlled-phase reductions for known computational-basis operands.
queryCount	9	Actual retained-circuit state/spin queries, including bounded report previews.
coherentOutputExecutions	1	Separate circuit executions actually performed to retain the coherent premeasurement output.

q.circuit run shots 1024 basis z

Actual native command answer

id	"3bfe5150b5254eeaa1039e7015acc782"
active	false
phase	"completed"
programName	"file_1e2ecb4b44f54154"
programHash	"9214cacc99b995dc5c4eb192894693b50b8f97e380bdc9dbf64750f01599e687"
qubits	2
initialBasis	"0"
basis	"z"
requestedShots	"1024"
completedShots	"1024"
measuredQubits	"2048"
counts	{"executedCircuitInstructions":"2050","backendGateApplications":"2050","actualAmplitudeWrites":"17420","actualTensorProducts":"4100","actualPreparedQubits":"2050","completedShots":"1024","measuredQubits":"2048","uniformDraws":"2048","exactCPReductions":"0","queryCount":"2050","coherentOutputExecutions":"1"}
counterStart	"8"
counterEnd	"2056"
error	""
checkpointError	""

Values above come from this command's returned result. A returned zero or false is data. Model verification below checks a separate procedural snapshot.

Original native product version	3.1.1-candidate
Recorded UTC	2026-10-10T19:13:32.1567993Z
Execution completion UTC	2026-10-10T19:13:32.9391154Z
Completion scope	Matched selected native sequence completion barrier observed at this UTC; this is execution evidence, not independent numerical verification.
Command elapsed (ms)	25.436400000000003
Saved procedural descriptor SHA-256	8851eb9b8dd83093c90097cf4c958108905e5bd1624bd3b1ad5c8a84cacc4117
Field seed	e1a83f29c3d5835b5b7f46bcf5334998f005abebc420be459e764fbeef6ae468
Measurement seed	1efd6137fe2bd4ff0be90691ad39eb2ec4dabcc886f34c2b08ef0283c9fd9949
Record provenance	Verified native archive snapshot: sequence d930f97c7e374b0b8d194b446cd640ab; stage completed.
Original input filename	gate-bell.csv
Original input format	csv-gates
Original input SHA-256	97ba88a3466d1fcdc787f12012b6e202a6450391418573209729d76cf8d0c03a
Selected native archive SHA-256	b3d82a1bbfce603b66f9df444c3d1d810691da03ed835055108c7e9e3076099b
Selected native archive stage	completed
Response data scope	Recorded native payload; chart scope is labeled per figure
Job phase	completed
Job elapsed (ms)	623.1196

programName	`file_1e2ecb4b44f54154`
programHash	`9214cacc99b995dc5c4eb192894693b50b8f97e380bdc9dbf64750f01599e687`
qubits	`2`
initialBasis	`0`
basis	`z`
phase	`completed`
requestedShots	`1024`
completedShots	`1024`
measuredQubits	`2048`
elapsedMs	`623.1196`
seed	`421d696fd15594faa8c7ed33687909240d8fe2f6e3aeccffa48705b305bcf05f`
counterStart	`8`
counterEnd	`2056`
error	`Not returned`
checkpointError	`Not returned`

Circuit gate sequence

Qubits / input basis	`2 / 0`



Layers 1-2; full exact angles are retained in the program descriptor.

Wires are indexed q0 through q1. Controls are dots; targets are labeled boxes. Gate angles below each layer are in radians. Diagram layers follow program order.

Exact gate operands

Layer / gate	Qubit indices	Angles (radians)
1. h	`[0]`	`[]`
2. cx	`[0,1]`	`[]`

Equation representations from recorded data

Symbolic declared circuit evolution from actual retained chronological program operands. Local matrices are native-convention formulas with recorded angle bindings; no full-register matrix, numerical state, probability, amplitude or extra shot is calculated by export. Histogram and spin figures remain separately recorded actual results.

Editable equation text

```
|psi_out> = U[2] U[1] |00>  Computational-basis index b = 0; register width n = 2; |b> = |00>.  Gate
rows are chronological; the first row acts first (rightmost in the operator product). Global
bitstrings have qubit0 at the right. Local two-operand matrices use |operand1,operand0> =
|00>,|01>,|10>,|11>: the first listed operand is the local least-significant bit, including a
controlled gate's control.  U[1] = H; operands = [0]; all = False; angle radians = []    Local
symbolic operator: 1/sqrt(2) * [["1","1"],["1","-1"]]  U[2] = CX; operands = [0,1]; all = False; angle
radians = []    Local symbolic operator:
[["1","0","0","0"],["0","0","0","1"],["0","0","1","0"],["0","1","0","0"]]  Born: P(s | basis) =
|<s;basis|psi_out>|^2 = |<s| B_basis |psi_out>|^2; B_basis is the measurement rotation (basis-ket
matrix adjoint): B_Z = I, B_X = tensor(H), B_Y = tensor(H*S_dagger). These are mathematical basis
equivalents, not extra executed gates. Recorded measurement basis: z  Local Bloch: rho_q =
partial_trace_other_qubits(|psi_out><psi_out|); x=Tr(rho_q X)=2Re(rho_q[0,1]); y=Tr(rho_q
Y)=-2Im(rho_q[0,1]); z=Tr(rho_q Z)=rho_q[0,0]-rho_q[1,1]. Entangled local states can have x=y=z=0.
```

Copy this text to an editable local draft. Editing a draft does not change the original CSV, native circuit or archived computation. The PDF text is selectable; the visual mathematics below is a presentation of the same symbols.

Conventional mathematical form

$$|\psi_{out}> = U_2 \quad U_1 \quad |00>$$

$$P(s \mid B) = |<s| B | \quad \psi_{out} >|^2$$

$$r_{q,k} = Tr(\rho_q \quad \sigma_k)$$

Operators act from right to left; U1 is the first archived gate instruction when any gates are present. B is the mathematical measurement rotation, equivalent to projecting onto the selected basis kets; it does not add gates executed by this exporter. Local reduced density matrices allow zero Bloch vectors for entangled mixed marginal states.

H - local mathematical operator

Basis / operand convention `|0>,|1>`

$$\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Matrix prefactor `1/sqrt(2)`

CX - local mathematical operator

Basis / operand convention

`|operand1,operand0> = |00>,|01>,|10>,|11>; operand0 is local least-significant bit and control where controlled. No global register matrix.`

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix}$$

Copyable conventional notation (LaTeX)

```
|\psi_{\mathrm{out}}\rangle=U_2U_1|b\rangle;\quad P(s\mid B)=|\langle s|B|\psi_{\mathrm{out}}\rangle|^2;\quad r_{q,k}=\mathrm{Tr}(\rho_q\sigma_k)
```

Histogram scope `Complete recorded outcome distribution`

Returned histogram shot mass `1024 / 1024`

Completed joint shots / requested



Observed joint outcomes - most frequent 24 shown



Bars show returned counts, not probability amplitudes. Exact rows follow; omitted preview rows are explicitly identified.

Circuit work accounting

Job-local actual work for this recorded circuit run, identified by its native job id and programHash. These counts remain separate from the saved circuit lifetime totals and from the procedural model's CPU/GPU verification counters.

executedCircuitInstructions	2050
backendGateApplications	2050
actualAmplitudeWrites	17420
actualTensorProducts	4100
actualPreparedQubits	2050
completedShots	1024
measuredQubits	2048
uniformDraws	2048
exactCPReductions	0
queryCount	2050
coherentOutputExecutions	1

Actual work includes independent shot replays and the separately retained coherent-output execution. Exact classical-input controlled-phase reductions are recorded. These units are not physical quantum gate throughput or FLOPS.

Retained circuit state before measurement

programHash	`9214cacc99b995dc5c4eb192894693b50b8f97e380bdc9dbf64750f01599e687`
capturedUtc	`2026-10-10T19:13:32.2557421Z`
premeasurement	`True`
selectedQubits	`2`
returnedQubits	`2`
omittedQubits	`0`
unrequestedQubits	`0`
qubitLimit	`4`
maximumFactorEntries	`65536`
scope	`Actual reduced Pauli expectations for the first at most four qubits of the retained coherent circuit output before independent measurement shots. Each queried coherent factor has at most 65536 stored entries. Omitted qubits were not evaluated and are not assigned zero vectors. This capture is separate from the saved procedural model and from postmeasurement shot states.`
verificationScope	`These are actual circuit-state calculations. The procedural-model q.verify CPU/GPU sampling gate does not independently verify these circuit snapshots.`

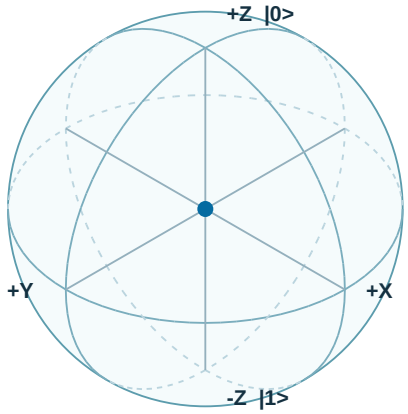
These local Pauli expectations were captured from this run’s actual coherent output before shot measurements. They are not reconstructed from the histogram, do not describe a later edited circuit, and are separate from procedural q.verify sampling.

Circuit local spin / qubit 0

Spin figure source	Selected coherent circuit premeasurement output; reduced single-qubit state.
qubit	0
premeasurement	True
stateCharacter	maximally-mixed-single-qubit
purity	0.5
blochLengthSquared	0
pure	False
mixed	True
exactlyZeroBloch	True
directionDefined	False
directionResolvedAtTolerance	False
directionTolerance	2E-09
spinScope	X/Y/Z are reduced Pauli expectations from the actual retained coherent register, not the separately saved procedural model or a postmeasurement shot. Entanglement can give a mixed local state. An exact zero vector has no preferred direction; a nonzero vector below tolerance retains its components and is labelled near-maximally mixed.

This is a retained coherent-state prediction, separate from observed shot counts and the saved procedural model checked by q.verify. A zero local vector has no preferred direction; |0> is the north-pole basis label, not a zero vector.

Bloch sphere and spin arrow



X spin 0.000000

Y spin 0.000000

Z spin 0.000000

Vector length = 0.000000

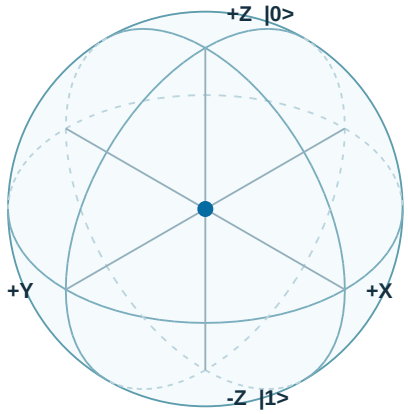
Figure: orthographic view of the reported X/Y/Z expectations.
Arrow length is preserved in 3D; projection may look shorter.
Zero vector: no preferred local direction; center marker, not |0>.
`Recorded XYZ = (0, 0, 0)`

Circuit local spin / qubit 1

Spin figure source	Selected coherent circuit premeasurement output; reduced single-qubit state.
qubit	1
premeasurement	True
stateCharacter	maximally-mixed-single-qubit
purity	0.5
blochLengthSquared	0
pure	False
mixed	True
exactlyZeroBloch	True
directionDefined	False
directionResolvedAtTolerance	False
directionTolerance	2E-09
spinScope	X/Y/Z are reduced Pauli expectations from the actual retained coherent register, not the separately saved procedural model or a postmeasurement shot. Entanglement can give a mixed local state. An exact zero vector has no preferred direction; a nonzero vector below tolerance retains its components and is labelled near-maximally mixed.

This is a retained coherent-state prediction, separate from observed shot counts and the saved procedural model checked by q.verify. A zero local vector has no preferred direction; |0> is the north-pole basis label, not a zero vector.

Bloch sphere and spin arrow



X spin 0.000000

Y spin 0.000000

Z spin 0.000000

Vector length = 0.000000

Figure: orthographic view of the reported X/Y/Z expectations.
Arrow length is preserved in 3D; projection may look shorter.
Zero vector: no preferred local direction; center marker, not |0>.
Recorded XYZ = (0, 0, 0)

Complete native histogram

Histogram rows included 2

Bitstring (qubit 0 at right)	Completed joint shots
00	526
11	498

Native interpretation and evidence boundaries

Completion meaning
A synchronous return or recorded final snapshot is shown. Command acceptance alone is not independent verification.

Documented method for the requested operation
Replay the selected circuit independently for each requested joint measurement shot using seeded Born sampling; a separate coherent-output execution is retained. Acceptance is not completion; matched job-local work and histogram fields describe actual execution.

Native method / convention
bitOrder=Most significant displayed bit at left; qubit 0 at right.; scope=Full joint bitstrings from independently replayed coherent registers and sequential collapsing Born measurements. Histograms contain completed shots only; partial shot work and consumed random draws are counted separately.

Native echoed operands
qubits=2; basis=z; initialBasis=0; requestedShots=1024

Complex values
[real, imaginary] stores both components of a complex number. An exactly returned [0,0] means zero amplitude/overlap; its phase is undefined. Probabilities are squared magnitudes, while density/operator entries are not individual probabilities.

Returned native checks
not returned

Validation boundary
Returned booleans/residuals describe native checks at the returned tolerance and scope. ok=true is a command return/acceptance flag. Independent verification is not established unless a matched native verification result and its coverage are explicitly returned; the exporter does not recompute or certify it.

Command-local work returned
completedShots=1024

Elapsed time scope
Native response elapsedMs=not returned; result elapsedMs=623.1196; result totalMs=not returned. These are returned command/job timing scopes, not export time or a measured operation rate.

Procedural model verification receipt

Receipt sampling
NATIVE SAMPLE VERIFICATION NOT REQUESTED

Verification phase
not-requested

Receipt release
Archived native return available; no sample verification inferred

The verification materializes sampled current-model predictions (amplitudes, spin and Born probabilities) and checks CPU / available GPU agreement. These are snapshot predictions, not measurement observations. Its PASS applies to this descriptor and sampled addresses; it does not independently certify arbitrary calculator outputs or every model address.

Folder receipt verification is not required for this command; inspect its own returned verification result if present.

The command result and returned checks are shown above. Complete accepted/final native fields, lifetime counters, hashes and verification rows remain in the matched TXT/JSON or Complete PDF view.

Recorded accounting

Scope: saved native status accompanying this recorded return. These values are not this command's work delta and are not guaranteed to describe the current live session. An acceptance snapshot can precede later job or per-file verification evidence; those results retain their own job-local counts.

A recorded 0 is a genuine zero in this snapshot. Not returned means the field was absent; missing work is never replaced with 0. Defining a circuit does not add execution counts. Represented credits, actual procedural calculations, circuit execution and selected-register work are separate quantities, not FLOPS or physical quantum throughput.

Inactive counter groups are compacted below when every returned value in that scope is exactly 0 or false. Mathematical results, nonzero values and unknown fields are not changed. Complete PDF, TXT and JSON retain the exact individual counter rows.

Procedural domain (descriptor, not allocation)

Counter	Recorded value	Meaning
gridM	20000000	Address-grid width in the saved procedural descriptor.
gridL	25000000	Address-grid height in the saved procedural descriptor.
stateCount	500000000000000	Number of addressable single-qubit family states; not allocated amplitudes, calculated states or jointly entangled qubits.
hasPrepared	False	Whether procedural preparation is committed in this saved model; independent of a circuit definition.
migrationT	0	Saved procedural preparation parameter; not the circuit's input state or run time.

Represented procedural credits

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Procedural model lifetime actual work

Inactive snapshot: all 7 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Selected tranche snapshot

Inactive snapshot: all 6 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Tranche lifetime actual work

Inactive snapshot: all 5 returned counters/flags in this scope are exactly 0 or false; 0 mapped fields were not returned. No nonzero activity was recorded in those returned fields. Exact individual values remain in Complete PDF, TXT and JSON.

Circuit lifetime actual work

Counter	Recorded value	Meaning
actualPreparedQubits	2059	Qubits actually prepared for circuit shot replays and retained coherent-output executions.
executedCircuitInstructions	2059	Stored program instructions actually traversed by circuit executions; not the definition’s gateCount.
backendGateApplications	2059	Actual local/joint backend gate calls; separate from program instruction count and exact CP reductions.
actualAmplitudeWrites	17472	Actual circuit backend amplitude writes; not physical gates or FLOPS.
actualTensorProducts	4100	Actual circuit backend tensor-product arithmetic work.
completedShots	1032	Completed joint circuit measurement shots across recorded jobs; not probabilities or requested shots.
measuredQubits	2056	Completed circuit qubit measurements, including work from partial shots.
uniformDraws	2056	Seeded random variates actually consumed by circuit measurements; a cancelled partial draw can still count.
exactCPReductions	0	Recorded exact controlled-phase reductions for known computational-basis operands.
queryCount	2059	Actual retained-circuit state/spin queries, including bounded report previews.
coherentOutputExecutions	2	Separate circuit executions actually performed to retain the coherent premeasurement output.